

Supplementary Material for “Nonlinear Tucker Decomposition with Polynomial Links: Rank Inflation and Tensor Recovery”

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This document provides supporting material for the accompanying paper. It is organized into additional model and algorithm details, complete proofs, and extended experimental results. Notation follows the main paper unless stated otherwise, with standard tensor conventions as in [1].

S1. ADDITIONAL MODEL AND ALGORITHM DETAILS

A. Detailed p -Fold Tucker Construction

Section III introduces the Tucker polynomial through p -fold terms. Here we give the explicit index construction and show how sharing the mode factors produces NTD-PL.

For a Tucker signal, a formal p -fold Tucker term may be written as

$$\tilde{Y}_{\mathbf{i}}^{(p)} = \sum_{\mathbf{r}^{(1)}, \dots, \mathbf{r}^{(p)}} \left(\prod_{\ell=1}^p X_{r_1^{(\ell)}, \dots, r_N^{(\ell)}} \right) \prod_{n=1}^N \mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)}. \quad (\text{S1})$$

For $p = 1$, this is the ordinary Tucker entry formula. For $p = 2$, each mode projection couples two Tucker index tuples, giving a quadratic Tucker term. More generally, (S1) is the tensor analogue of augmenting a linear map with polynomial interactions among Tucker coordinates.

For fixed mode maps, each output entry is a degree- p polynomial in the core entries. When $N = 1$, this is an ordinary vector-valued homogeneous polynomial. For $N > 1$, the coefficient map is separated across the output modes in Tucker form.

The general construction is not the parameterization used by the estimator: each mode projection $\mathcal{A}^{(n,p)}$ has $I_n R_n^p$ entries and a separate set of such projections would be needed for every active degree. NTD-PL keeps the same interaction pattern but shares the original Tucker factors across orders,

$$\mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)} = \prod_{\ell=1}^p \mathcal{A}_{i_n, r_n^{(\ell)}}^{(n)}. \quad (\text{S2})$$

Substituting (S2) into (S1) gives exactly $\mathcal{S}^{\odot p}$, where $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$. Thus a degree- p NTD-PL term is a shared-factor p -fold Tucker term: it couples p Tucker index tuples while reusing the same mode factors.

This viewpoint explains why the model can increase effective rank without storing an untied high-order tensor block. The additional trainable variables are the response coefficients, while the high-order features are induced by repeated use of the same Tucker factors.

B. Projection Interpretation of the Response Diagnostic

The diagnostic D_{link} in (23) can be interpreted as a projection test for residual structure that depends on the current Tucker estimate. Let $\Phi_{\Omega, J}$ be the response feature matrix built from a fixed Tucker reconstruction $\mathcal{S}_T$ and let $\mathbf{P}_J$ denote the orthogonal projector onto its column space. When $\lambda_\beta = 0$, the refreshed-response residual satisfies

$$R_{\text{resp}} = \|(\mathbf{I} - \mathbf{P}_J)y_\Omega\|_2^2. \quad (\text{S3})$$

Thus, D_{link} is the residual-energy ratio, in dB, between the identity response and the projection onto the response-feature subspace generated by the fixed Tucker fit. With ridge regularization, the response coefficients are obtained from the corresponding shrinkage operator, while R_{resp} still denotes the data residual without the ridge penalty. A large positive D_{link} means that the residual is aligned with scalar functions of the current Tucker estimate, whereas a small or nonpositive D_{link} suggests increasing multilinear rank, changing the tensor prior, or revisiting the observation model.

C. Masked-Entry Generalization

Let $\Theta \subset \mathbb{R}^d$ be a bounded parameter set for a degree- P NTD-PL model, where

$$d = \prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P + 1). \quad (\text{S4})$$

For $\theta \in \Theta$, define the all-entry risk $\mathcal{L}(\theta) = M^{-1} \sum_{\mathbf{i}} \ell_{\mathbf{i}}(\theta)$, $M = \prod_n I_n$, and the masked empirical risk $\hat{\mathcal{L}}_m(\theta) = m^{-1} \sum_{t=1}^m \ell_{\mathbf{i}_t}(\theta)$, where the entries $\mathbf{i}_t$ are sampled uniformly.

Theorem S1 (Masked-entry uniform convergence). *Assume $0 \leq \ell_{\mathbf{i}}(\theta) \leq B_\ell$ and $\ell_{\mathbf{i}}(\theta)$ is G -Lipschitz in θ , uniformly over $\mathbf{i}$. If Θ lies in a Euclidean ball of radius B_Θ , then, with probability at least $1 - \delta$,*

$$\sup_{\theta \in \Theta} \left| \mathcal{L}(\theta) - \hat{\mathcal{L}}_m(\theta) \right| \leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}} \quad (\text{S5})$$

for every $0 < \varepsilon \leq B_\Theta$.

This standard covering bound identifies the complexity entering masked recovery: the dimension is the shared-factor parameter count, not that of an untied Tucker model with Veronese ranks.

D. Detailed Computational Scaling

Let $M = \prod_{n=1}^N I_n$, $R_\Pi = \prod_{n=1}^N R_n$, and $d_{\text{Tucker}} = R_\Pi + \sum_{n=1}^N I_n R_n$. Storing the model requires

$$d_{\text{NTD}} = d_{\text{Tucker}} + (J + 1) \quad (\text{S6})$$

parameters. For the power basis, $J = P$. A higher-rank Tucker model that explicitly matches the degree- P Veronese lift may require mode ranks $D_n = \binom{R_n+P-1}{P}$, with core size $\prod_n D_n$ and factor size $\sum_n I_n D_n$. NTD-PL avoids storing this larger Tucker tensor.

Response Refresh forms $\Phi_{\Omega,J}^T \Phi_{\Omega,J}$ and solves a small system, costing $\mathcal{O}(mJ^2 + J^3)$. Dense Tucker reconstruction and reverse contractions scale as

$$\mathcal{O}\left(MR_\Pi + \sum_{n=1}^N I_n R_n R_\Pi / R_n\right) \quad (\text{S7})$$

up to contraction-order constants, followed by $\mathcal{O}(MJ)$ work for the entrywise response. In masked settings, response and residual computations use m observed entries. Since J is small, the response update is typically minor relative to the Tucker contractions.

S2. PROOFS

A. Proof of Proposition 1

Substitute (7) into (5). The products can be regrouped as

$$\begin{aligned} [\mathcal{T}_p(\mathcal{X})]_{\mathbf{i}} &= \prod_{\ell=1}^p \left(\sum_{\mathbf{r}^{(\ell)}} X_{\mathbf{r}^{(\ell)}} \prod_{n=1}^N A_{i_n, r_n^{(\ell)}}^{(n)} \right) \\ &= S_{\mathbf{i}}^p. \end{aligned}$$

Applying this identity entrywise gives $\mathcal{T}_p(\mathcal{X}) = \mathcal{S}^{\odot p}$.

B. Proof of Proposition 2

Let

$$S_{\mathbf{i}} = \sum_{r_1, \dots, r_N} X_{r_1, \dots, r_N} \prod_{n=1}^N A_{i_n, r_n}^{(n)}.$$

Expanding $S_{\mathbf{i}}^p$ gives products of p Tucker index tuples. For each mode n , the resulting mode-dependent terms are all degree- p monomials in the row $A_{i_n, \cdot}^{(n)}$. These monomials are indexed by multi-indices $\alpha \in \mathbb{N}^{R_n}$ with $|\alpha| = p$, of which there are $\binom{R_n+p-1}{p}$. Collecting identical monomials yields a Tucker representation of $\mathcal{S}^{\odot p}$ whose mode- n factor is the degree- p Veronese lift of $\mathbf{A}^{(n)}$. This proves the stated rank bound.

C. Proof of Theorem 1

We first record a standard fact about Veronese evaluations. Let $D = \binom{R+p-1}{p}$. There exists a matrix $\mathbf{A} \in \mathbb{R}^{D \times R}$ whose degree- p Veronese feature matrix has rank D . Indeed, the homogeneous degree- p monomials in R variables are linearly independent polynomials, so one can choose D evaluation points for which the square evaluation matrix is nonsingular. Consequently, for $I \geq D$, a generic $\mathbf{A} \in \mathbb{R}^{I \times R}$ has a degree- p Veronese lift with full column rank D .

Choose generic factor matrices $\mathbf{A}^{(n)} = [a_1^{(n)}, \dots, a_R^{(n)}] \in \mathbb{R}^{I_n \times R}$ and define the CP-diagonal Tucker signal

$$\mathcal{S} = \sum_{r=1}^R a_r^{(1)} \otimes \dots \otimes a_r^{(N)}.$$

For generic factors, each mode unfolding of $\mathcal{S}$ has rank R , so the signal has exact Tucker rank $(R, \dots, R)$.

Now set the degree- p response coefficient to one and all other coefficients to zero, so $\mathcal{Y} = \mathcal{S}^{\odot p}$. Expanding entrywise gives

$$Y_{\mathbf{i}} = \left(\sum_{r=1}^R \prod_{n=1}^N A_{i_n, r}^{(n)} \right)^p = \sum_{|\alpha|=p} c_\alpha \prod_{n=1}^N \prod_{r=1}^R \left(A_{i_n, r}^{(n)} \right)^{\alpha_r},$$

where $c_\alpha = p! / (\alpha_1! \dots \alpha_R!) > 0$. Let $\mathbf{B}^{(n)} = \mathbf{A}^{(n)\langle p \rangle} \in \mathbb{R}^{I_n \times D}$ be the Veronese lift. By the preceding full-rank fact, each $\mathbf{B}^{(n)}$ has column rank D for generic $\mathbf{A}^{(n)}$. The mode- n unfolding of $\mathcal{Y}$ factors as $\mathbf{B}^{(n)}$ times a full-column-rank right factor built from the remaining lifted factors, with a nonsingular diagonal scaling by the positive coefficients c_α . Therefore every mode unfolding has rank D . The exact standard Tucker rank of $\mathcal{Y}$ is $(D, \dots, D)$, while $\mathcal{Y}$ is represented by a degree- p NTD-PL model using the rank- $(R, \dots, R)$ Tucker signal.

D. Proof of Corollary 1

The direct parameterization in (5) stores $I_n R_n^p$ entries in the mode- n map. By Proposition 2, the shared degree- p term can also be written as a Tucker tensor with ranks at most $D_n(p) = \binom{R_n+p-1}{p}$. A standalone Tucker parameterization has $\prod_n D_n(p)$ core entries and $\sum_n I_n D_n(p)$ factor entries. In contrast, NTD-PL with the power basis reuses the core and factors of $\mathcal{S}$ and adds only one response coefficient per polynomial term. Summing degrees up to P adds $P + 1$ response coefficients, proving the stated comparison.

E. Proof of Proposition 3

If $g(s) = \sum_{p=0}^P \gamma_p s^p$ and $\mathcal{Y} = g(\mathcal{S}^*)$, choose the Tucker component of NTD-PL to be the Tucker representation of $\mathcal{S}^*$ and set $\beta_p = \gamma_p$ for all $p = 0, \dots, P$. Then $f_{\beta}(\mathcal{S}^*) = g(\mathcal{S}^*) = \mathcal{Y}$, giving exact representation with the same Tucker rank.

F. Proof of Theorem 2

By the Weierstrass approximation theorem, for every $\epsilon > 0$ there exists a polynomial $\pi_P(s) = \sum_{p=0}^P \beta_p s^p$ such that $\sup_{s \in [a, b]} |f(s) - \pi_P(s)| \leq \epsilon$. Every entry of $\mathcal{S}^*$ lies in $[a, b]$, so each entrywise approximation error is bounded by ϵ . Therefore

$$\|f(\mathcal{S}^*) - \pi_P(\mathcal{S}^*)\|_F^2 = \sum_{\mathbf{i}} |f(S_{\mathbf{i}}^*) - \pi_P(S_{\mathbf{i}}^*)|^2 \leq M \epsilon^2,$$

which gives the stated bound after taking square roots.

G. Proof of Proposition 4

For a fixed Tucker signal, Response Refresh solves the β -subproblem exactly, hence it cannot increase $\mathcal{F}_J$. The subsequent Tucker update is assumed to satisfy the sufficient decrease condition (28), and therefore also cannot increase $\mathcal{F}_J$. Combining the two stages gives monotone decrease over the outer iteration. Since $\mathcal{F}_J$ is bounded below by zero when the regularization weights are nonnegative, summing (28) over k gives

$$c \sum_{k=0}^K \|\theta^{k+1} - \theta^k\|_2^2 \leq \mathcal{F}_J(\theta^0) - \mathcal{F}_J(\theta^{K+1}) \leq \mathcal{F}_J(\theta^0).$$

Letting $K \rightarrow \infty$ proves square summability of the steps.

H. Proof of Theorem 3

For fixed J , the objective is a polynomial function of the Tucker parameters and response coefficients when the monomial basis is used. For any other fixed polynomial basis spanning the same space, it remains polynomial after a linear change of coefficients. The linear-spline basis is piecewise polynomial and semi-algebraic; standard RBF bases are analytic and belong to the standard tame classes covered by KL convergence theory. With standard box or normalization constraints used to fix the Tucker gauge, the corresponding constrained objective satisfies the Kurdyka–Lojasiewicz property. Assumption 1 gives sufficient decrease and a relative error bound. The standard KL convergence theorem for nonconvex descent sequences then implies finite length of the iterates and convergence to a critical point of $\mathcal{F}_J$.

I. Proof of Theorem S1

Fix $0 < \varepsilon \leq B_\Theta$, and let $\mathcal{N}_\varepsilon$ be an ε -net of Θ in Euclidean norm. Since Θ lies in a d -dimensional ball of radius B_Θ , it admits a net with

$$|\mathcal{N}_\varepsilon| \leq \left(\frac{3B_\Theta}{\varepsilon} \right)^d.$$

For any fixed $\theta_0 \in \mathcal{N}_\varepsilon$, the random variables $\ell_{i_t}(\theta_0)$ are independent and bounded in $[0, B_\ell]$. Hoeffding's inequality gives

$$\Pr \left(\left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \geq u \right) \leq 2 \exp \left(-\frac{2mu^2}{B_\ell^2} \right).$$

A union bound over $\mathcal{N}_\varepsilon$ shows that, with probability at least $1 - \delta$,

$$\max_{\theta_0 \in \mathcal{N}_\varepsilon} \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \leq B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}.$$

For an arbitrary $\theta \in \Theta$, choose $\theta_0 \in \mathcal{N}_\varepsilon$ with $\|\theta - \theta_0\|_2 \leq \varepsilon$. The uniform Lipschitz assumption implies

$$|\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| \leq G\varepsilon, \quad |\widehat{\mathcal{L}}_m(\theta) - \widehat{\mathcal{L}}_m(\theta_0)| \leq G\varepsilon.$$

Combining these two approximation errors with the net bound and taking the supremum over Θ gives the stated inequality.

S3. ADDITIONAL EXPERIMENTAL DETAILS

A. Controlled Nonlinear Tensors

The controlled tensors in Fig. 3 have size $10 \times 10 \times 10$ and Tucker rank $(4, 4, 4)$. For each of ten seeds, all entries of the core and factors are drawn independently from $\mathcal{N}(0, 1)$. Suppressing the seed index, we form

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \mathbf{A}^{(2)}, \mathbf{A}^{(3)} \rrbracket. \quad (\text{S8})$$

We then rescale $\mathcal{S}$ to unit root-mean-square value, keeping the same notation. The nonlinear panels apply one of the following scalar functions entrywise:

$$g(s) \in \left\{ s^2 + s^3, \tanh(\kappa_{\tanh} s), \frac{e^{\kappa_{\exp} s} - 1}{\kappa_{\exp}} \right\}. \quad (\text{S9})$$

Writing $q = q_{0.9}(|\mathcal{S}|) + 10^{-8}$, where $q_{0.9}$ is the empirical 90th percentile, we set $\kappa_{\tanh} = 1.5/q$ and $\kappa_{\exp} = 1/q$.

To isolate the nonlinear part, we remove the component of $g(\mathcal{S})$ parallel to $\mathcal{S}$:

$$\mathcal{R} = g(\mathcal{S}) - \frac{\langle g(\mathcal{S}), \mathcal{S} \rangle}{\|\mathcal{S}\|_F^2 + 10^{-8}} \mathcal{S}. \quad (\text{S10})$$

No intercept is included in this projection. We also rescale $\mathcal{R}$ to unit root-mean-square value, again keeping the same notation. Consequently, $\mathcal{S}$ and $\mathcal{R}$ are orthogonal and have the same energy. The clean nonlinear measurement is

$$\mathcal{Y}_\alpha^{\text{clean}} = \sqrt{1 - \alpha} \mathcal{S} + \sqrt{\alpha} \mathcal{R}. \quad (\text{S11})$$

We use $\alpha \in \{0.10, 0.15, \dots, 0.40\}$, so α is exactly the energy fraction assigned to the nonlinear residual. The identity panel instead uses $\mathcal{Y}_\alpha^{\text{clean}} = \mathcal{S}$ for every α .

Finally,

$$\mathcal{Y}_\alpha = \mathcal{Y}_\alpha^{\text{clean}} + \mathcal{E}, \quad E_i \sim \mathcal{N}(0, 10^{-3}), \quad (\text{S12})$$

which gives 30 dB SNR because every clean tensor has unit root-mean-square value. Signal seeds are $0, \dots, 9$; noise seeds are deterministically derived from the signal seed, response, and α in the released experiment script.

All 1000 entries are used for both fitting and evaluation. Tucker uses rank $(4, 4, 4)$, SVD initialization, at most 1000 HOOI iterations, and tolerance 10^{-4} . The matched parameter choices are CP rank 6 and TT rank $(1, 3, 3, 1)$. CP uses random initialization with the corresponding signal seed, at most 1000 ALS iterations, and tolerance 10^{-8} ; TT uses the standard TT-SVD. NTD-PL uses Tucker rank $(4, 4, 4)$ and $P = 6$ in the α sweep. It starts from a Tucker fit with at most 50 iterations and then runs 1000 Adam iterations with core and factor learning rates 10^{-4} and 3×10^{-4} . The core, factor, and response regularization weights are all 10^{-6} , the response coefficients are refreshed every five iterations, and polynomial degrees are activated by continuation. These settings give 184, 186, 150, and 191 parameters for Tucker, CP, TT, and NTD-PL, respectively.

Figure S1 fixes $\alpha = 0.25$ and refits NTD-PL for every $P \in \{1, \dots, 6\}$ using the same ten seeds. Figure S2 also fixes $\alpha = 0.25$ and $P = 6$; its curves are the mean RMSE over the ten seeds, with shaded bands showing one standard deviation.

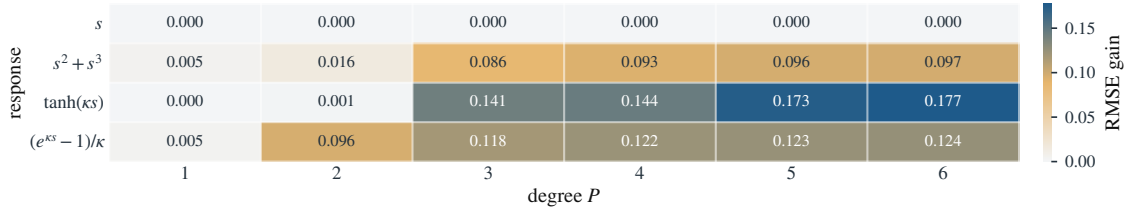


Fig. S1. Degree-dependent controlled gains under the same 30 dB noisy generator as Fig. 3. Entries report the RMSE reduction of NTD-PL relative to matched-rank Tucker. The identity response s is a linear control; gains appear when the active polynomial degree can approximate the nonlinear response.

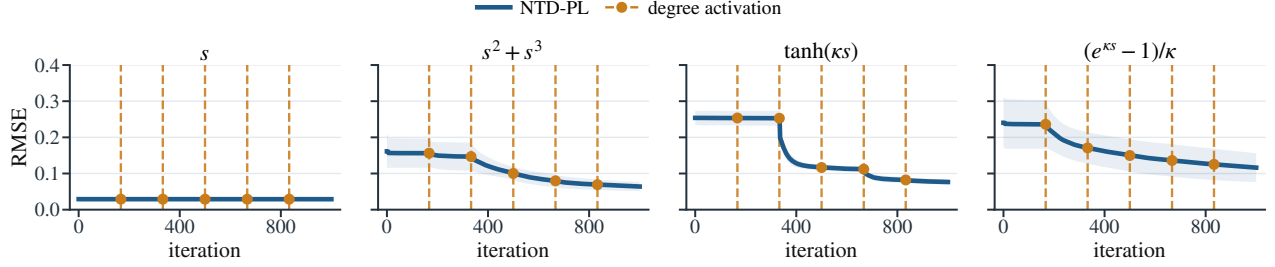


Fig. S2. NTD-PL optimization traces at $\alpha = 0.25$ and $P = 6$. Curves show mean RMSE over ten seeds and shaded bands show one standard deviation. Vertical markers indicate the activation of degrees 2–6 at iterations 167, 333, 500, 667, and 833.

B. Metrics and Normalization

For a test index set Γ , the RMSE is

$$\text{RMSE} = \left(|\Gamma|^{-1} \sum_{i \in \Gamma} (Y_i - \hat{Y}_i)^2 \right)^{1/2}. \quad (\text{S13})$$

For completion, Γ contains the entries hidden during fitting. For reconstruction, Γ contains all tensor entries. We also report

$$\text{NMSE} = \frac{\sum_{i \in \Gamma} (Y_i - \hat{Y}_i)^2}{\sum_{i \in \Gamma} Y_i^2}, \quad (\text{S14})$$

$$\text{NMSE(dB)} = 10 \log_{10}(\text{NMSE}).$$

Table I computes NMSE(dB) separately for each CAVE scene and then reports the mean and standard deviation over the 15 scene values. Spectral angle mapper is computed pixelwise on spectral vectors,

$$\text{SAM}(\mathbf{y}, \hat{\mathbf{y}}) = \arccos \frac{\langle \mathbf{y}, \hat{\mathbf{y}} \rangle}{\|\mathbf{y}\|_2 \|\hat{\mathbf{y}}\|_2 + \epsilon}, \quad (\text{S15})$$

and then averaged over valid pixels. The small ϵ is used only to avoid division by zero for empty or near-empty spectra. Before fitting, each hyperspectral cube is divided by its maximum entry. All compared methods use the same normalized cube and the same train/test masks.

C. Hyperspectral Protocol

The CAVE multispectral image database [2], [3] is used for the primary hyperspectral experiments. Reconstruction uses all entries for fitting and evaluation. In the noise-robustness experiment, Gaussian noise is added before fitting and metrics are computed against the clean cube. Completion samples observation masks at the requested missing rates and evaluates RMSE* and SAM* on the entries hidden during fitting. Tucker and NTD-PL use identical Tucker ranks; the default NTD-PL

instance adds only the polynomial response coefficients and the associated training schedule. The reconstruction comparison in Table I uses Tucker ranks (12, 12, 4), (24, 24, 4), and (36, 36, 4), corresponding to approximately 13k, 27k, and 42k parameters. At these budgets, the matched CP ranks are 12, 26, and 40, and the TT ranks are (1, 5, 5, 1), (1, 7, 7, 1), and (1, 9, 9, 1), respectively. At every budget, NTD-PL is refitted independently for $P = 2, 4$, and 6.

The completion experiment uses Tucker rank (33, 33, 4), $P = 4$, and 300 optimization iterations for both methods at missing rates 0.1, 0.3, 0.5, and 0.7.

The noise-robustness evaluation uses all 15 CAVE scenes at their native $512 \times 512 \times 31$ size with rank (12, 12, 6) and $P = 4$. Gaussian noise is added after global max normalization at SNR levels 40, 30, 20, and 10 dB; the clean case uses the same protocol without added noise. The clean evaluation tensor is shared by Tucker and NTD-PL for each scene, and both methods use 100 optimization iterations. Across the tested noise levels, NTD-PL reduces RMSE by 6.16–6.20% and SAM by 4.79–5.82% relative to matched-rank Tucker.

External hyperspectral experiments use Jasper Ridge ($100 \times 100 \times 198$), Samson ($95 \times 95 \times 156$), and Urban ($307 \times 307 \times 162$) after the same global normalization. Their Tucker ranks are (11, 11, 43), (10, 10, 35), and (43, 43, 47), respectively. NTD-PL uses $P = 4$, and both methods use 300 optimization iterations. Table S2 reports the corresponding reconstruction results. At the same Tucker rank, NTD-PL improves RMSE by 3.6–7.9% and also improves SAM on all three cubes. The NMSE changes are positive as well, with gains of 0.32–0.72 dB. These results serve as a supplementary cross-dataset check beyond the CAVE acquisition setting.

The higher-rank Tucker comparison in Table II searches over higher Tucker ranks and reports the first Tucker model whose RMSE matches the lower-rank NTD-PL reference with $P = 6$. Its purpose is to quantify the parameter increase

TABLE S1

CAVE RECONSTRUCTION WITH OBSERVATION NOISE OVER ALL 15 SCENES. MODELS ARE FITTED TO CLEAN OR NOISY TENSORS, AND METRICS ARE COMPUTED AGAINST THE CLEAN TENSOR; GAINS ARE RELATIVE ERROR REDUCTIONS OVER MATCHED-RANK TUCKER.

SNR	Clean RMSE↓			Clean SAM↓		
	Tucker	NTD-PL	Rel. gain	Tucker	NTD-PL	Rel. gain
Clean	0.0319 ± 0.0268	0.0299 ± 0.0257	6.16%	14.02 ± 7.09	13.35 ± 6.47	4.79%
40 dB	0.0319 ± 0.0268	0.0299 ± 0.0257	6.16%	14.02 ± 7.09	13.35 ± 6.47	4.79%
30 dB	0.0319 ± 0.0268	0.0299 ± 0.0257	6.16%	14.02 ± 7.09	13.35 ± 6.47	4.82%
20 dB	0.0319 ± 0.0268	0.0299 ± 0.0257	6.19%	14.10 ± 7.09	13.39 ± 6.48	4.98%
10 dB	0.0320 ± 0.0268	0.0300 ± 0.0257	6.20%	14.75 ± 7.38	13.90 ± 6.79	5.82%

TABLE S2

EXTERNAL HYPERSPECTRAL RECONSTRUCTION. NTD-PL USES THE SAME TUCKER RANK AS THE TUCKER BASELINE AND ADDS ONLY THE POLYNOMIAL RESPONSE COEFFICIENTS.

Dataset	RMSE↓			SAM↓		
	Tucker	NTD-PL	Rel. gain	Tucker	NTD-PL	Rel. gain
Jasper Ridge	0.0462	0.0430	6.9%	9.14	8.16	10.7%
Samson	0.0263	0.0242	7.9%	5.70	5.65	1.0%
Urban	0.0444	0.0428	3.6%	8.23	7.94	3.5%

required by increasing ordinary Tucker rank when the residual can be explained by the learned response.

D. Mechanism Analysis Details

The mechanism analysis in Figs. 4 and 5 uses the $512 \times 512 \times 31$ CAVE scenes at rank (12,12,4) and sweeps the maximum power degree P . The rank-inflation summary in Table S3 reports the degree-wise gain, nonlinear-response rank, effective-rank increase, and tail capture. Figure 6 in the main paper visualizes the observed-domain error regimes, and Table S4 reports the spectral breakdown.

For the rank-inflation calculation, we fit a separate Tucker baseline and NTD-PL model for each scene and each $P \in \{2, 3, 4, 5, 6\}$, using all entries of the globally max-normalized cube. The Tucker rank is (12, 12, 4), the random seed is 0, and the NTD-PL runs use Tucker initialization, the power basis, degree continuation, and 120 optimization iterations. If $\hat{\mathcal{S}}$ is the tensor reconstructed from the NTD-PL Tucker parameters and $\hat{\mathcal{Y}}$ is its response-linked prediction, the nonlinear response component is

$$\hat{\mathcal{R}}_{\text{nl}} = \hat{\mathcal{Y}} - \hat{\beta}_0 \mathbf{1} - \hat{\beta}_1 \hat{\mathcal{S}} = \sum_{p=2}^P \hat{\beta}_p \hat{\mathcal{S}}^{\odot p}.$$

For a tensor $\mathcal{Z}$ and mode n , define

$$r_{.999}(\mathcal{Z}; n) = \min \left\{ k : \frac{\sum_{j=1}^k \sigma_j^2(\mathcal{Z}_{(n)})}{\sum_j \sigma_j^2(\mathcal{Z}_{(n)})} \geq 0.999 \right\},$$

where $\mathcal{Z}_{(n)}$ is the mode- n unfolding. Fig. 4 uses modes 1 and 2 and reports the median over the 30 scene-mode values for each annotation. The effective-rank increase is computed as $r_{.999}(\hat{\mathcal{Y}}; n) - r_{.999}(\hat{\mathcal{S}}; n)$. The RMSE gain is computed per scene as

$$g_{\text{RMSE}} = 100 \frac{\text{RMSE}_{\text{Tucker}} - \text{RMSE}_{\text{NTD-PL}}}{\text{RMSE}_{\text{Tucker}}},$$

TABLE S3

RANK-INFLATION SUMMARY ON CAVE SCENES AT RANK (12, 12, 4). GAINS ARE RELATIVE TO MATCHED-RANK TUCKER.

P	RMSE gain (%)	Nonlin. rank	Rank inc.	Tail cap.
2	2.82	14.0	0.0	0.009
3	6.15	16.5	2.0	0.061
4	7.65	18.5	3.5	0.104
5	8.22	19.5	4.5	0.115
6	8.79	19.5	5.0	0.127

TABLE S4

SPECTRAL-REGIME ERROR ANALYSIS ON CAVE SCENES AT RANK (12, 12, 4).

Regime	Pooled gain (%)	Median gain (%)	MSE contrib. (%)
400–500 nm	3.54	8.16	14.5
510–600 nm	7.60	10.50	43.5
610–700 nm	7.68	9.11	42.0

then averaged over scenes. For the reported tail-capture correlation, tail energy is the fraction of unfolding energy after the prescribed spatial Tucker rank 12. We divide the prediction tail energy by the corresponding measured-tensor tail energy, average modes 1 and 2 within each scene, and compute Spearman correlation with RMSE gain over all 75 scene-degree pairs.

For the response-curve calculation, we refit NTD-PL independently for each scene and each degree with the same rank, normalization, seed, and power basis; these runs use 180 optimization iterations. For the fitted signal values s_i , we evaluate $f_{\beta}(s_i)$ at 101 empirical quantiles from the 1st to the 99th percentile. Each scene curve is endpoint-normalized as

$$x_i = \frac{s_i - s_{.01}}{s_{.99} - s_{.01}}, \quad y_i = \frac{f_{\beta}(s_i) - f_{\beta}(s_{.01})}{f_{\beta}(s_{.99}) - f_{\beta}(s_{.01})}.$$

The plotted dark curve is the pointwise mean over the 15 scene curves. To compute the curvature label, let $\Delta s_i = s_{i+1} - s_i$ and $d_i = (f_{\beta}(s_{i+1}) - f_{\beta}(s_i)) / \Delta s_i$. For each scene we use

$$\text{Curv.} = \frac{1}{q_{.95}(|d|)} \frac{1}{m-2} \sum_{i=1}^{m-2} \left| \frac{d_{i+1} - d_i}{0.5(\Delta s_i + \Delta s_{i+1})} \right|,$$

where $m = 101$ and $q_{.95}(|d|)$ is the 95th percentile of the absolute local slopes. The value shown in each panel is averaged over scenes. The gain in each panel is relative to the mean RMSE at $P = 1$.

TABLE S5
CROSS-DOMAIN DATA CONSTRUCTION AND PREPROCESSING FOR THE RESPONSE DIAGNOSTIC.

Dataset	Units	Tensor unit	Selection and preprocessing	Tensor size	Tucker rank
CAVE	15	one scene	All 15 scenes at their native spatial and spectral resolution.	$512 \times 512 \times 31$	(24, 24, 4)
CBSD68	68	one RGB image	All 68 images; each image is resized to 96×96 .	$96 \times 96 \times 3$	(32, 32, 3)
CIFAR-10	1000	one RGB image	Class-balanced subset of the 10,000 test images; native resolution is retained.	$32 \times 32 \times 3$	(20, 20, 3)
COIL-100	100	one object-view tensor	All 100 objects; 36 azimuths from 0° to 350° in 10° steps; each view is resized from 128×128 to 48×48 .	$36 \times 48 \times 48 \times 3$	(8, 12, 12, 2)
smallNORB	25	one instance-view tensor	Training instances at elevation 4 and lighting 0; all 18 azimuths and both stereo views; images are resized from 96×96 to 64×64 .	$18 \times 64 \times 64 \times 2$	(12, 24, 24, 2)
KTH-Action	240	one annotated segment	240 of 2391 annotated action segments; 24 frames are sampled from each segment and resized to 72×96 RGB.	$24 \times 72 \times 96 \times 3$	(12, 18, 18, 3)
UCF101	240	one video clip	240 of 10,055 training clips; 24 frames are sampled from each clip and resized to 72×96 RGB.	$24 \times 72 \times 96 \times 3$	(12, 18, 18, 3)

TABLE S6
DATASET-LEVEL RESPONSE DIAGNOSTIC STATISTICS. EACH LOW, MIDDLE, OR HIGH ENTRY REPORTS MEAN D_{link} (dB) / MEAN NTD-PL RMSE GAIN (%) AFTER SORTING UNITS BY D_{link} WITHIN EACH DATASET.

Dataset	Units	ρ_s	Mean D_{link} (dB) / RMSE gain (%)		
			Low	Middle	High
CAVE	15	0.81	0.060 / 5.65	0.108 / 11.68	0.230 / 16.19
CBSD68	68	0.85	0.025 / 4.17	0.095 / 8.40	0.293 / 14.60
CIFAR-10	1000	0.93	-0.001 / -0.01	0.009 / 2.59	0.041 / 7.85
COIL-100	100	0.93	0.080 / 2.77	0.210 / 5.46	0.439 / 9.14
smallNORB	25	0.95	0.134 / 6.92	0.267 / 9.91	0.541 / 14.16
KTH-Action	240	0.88	0.049 / 2.74	0.108 / 4.27	0.274 / 10.87
UCF101	240	0.89	0.024 / 1.44	0.064 / 2.97	0.176 / 5.89

E. Response Diagnostic Details

The diagnostic D_{link} is computed from a fixed Tucker fit. First, a matched-rank Tucker model is trained under the same mask and normalization as the main experiment. Second, the Tucker reconstruction is held fixed and the response coefficients are refit by the ridge update in (16). The diagnostic is the residual-energy ratio before and after this response-only fit, reported in dB as defined in (23). This procedure is cheaper than joint NTD-PL training because it does not update the Tucker core and factors through the nonlinear response.

For the CAVE diagnostic row, each scene is treated as one reconstruction task. For the remaining rows, each image, object, or video clip is treated as one independent tensor unit. In all cases, D_{link} is computed before joint NTD-PL training. Units are sorted by D_{link} within each dataset and divided into three groups of nearly equal size; the table reports the mean diagnostic value and the eventual RMSE reduction in each group.

F. Cross-Domain Diagnostic Data

Table S5 summarizes the tensor units and preprocessing used for the natural-image [4], [5], object-view [6], [7], and action-video [8], [9] diagnostics.

Each tensor unit is max-normalized and fitted independently; no cross-unit tensor is formed. Diagnostic groups are assigned from D_{link} , not from the eventual NTD-PL gain.

G. Compute Resources

The runs use a CPU-only NumPy/SciPy/TensorLy implementation under Python 3.13 and Windows 11. Timings were

collected on 32 GB workstations with an Intel Core i5-13400F or i7-14700F processor. Independent jobs were parallelized with BLAS/OpenMP threads capped per worker; no GPU kernels were used. The CAVE noise sweep used 12 workers on the i7-14700F system.

Controlled synthetic jobs took less than 1 s per configuration. For CAVE reconstruction, Tucker took 10–17 s and NTD-PL 172–217 s per scene; random completion took 168–258 s and 463–1019 s, respectively, per scene and mask configuration. The CAVE response diagnostic took 7.0–9.6 s per scene. With 6 workers, the non-HSI image/object batch took about 77 s and each action-video batch took 5–6 min. The noise sweep required 9.7–32.6 s per Tucker fit and 61.0–167.7 s per NTD-PL fit; its 150 fits completed in about 21 min. Peak memory was not instrumented, but all runs completed within 32 GB RAM.

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